



Standard Operating Procedure (SOP)

Data Stewardship Management Procedure

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1. INTRODUCTION

1.1 OBJECTIVE

This process aims to establish principles, guidelines and standard procedures for the organization's data management in a systematic and controlled manner, ensuring that data remains accurate, secure, high quality and compliant with applicable laws and organizational standards. It also defines the roles and responsibilities of the Data Owner, Data Steward, Data Custodian (IT & XPO) and Data User to support the organization's Data Governance program and the Data & AI Transformation Framework.

1.2 SCOPE

This SOP applies to key business data under the organization's responsibility, referring but not limited to Customer, tenant, employee and business partner. For Key defined scopes are;

- 1) Data Creation and Ingestion
- 2) Data Definition and Business Meaning
- 3) Data Access Control (Role Based Access Control) matrix
- 4) Data Quality Management practices
- 5) Data Sharing and Data Reuse
- 6) Data Retention, Archiving, and Disposal

1.3 INTERNAL CONTROL FRAMEWORK

This process establishes data governance and control principles to support the appropriate, standardized, and lawful use of business data, access rights management, data quality assurance and data protection in compliance with applicable laws and regulations. This process plays a critical role in mitigating various risks, including but not limited to:

- Risks of unauthorized access or excessive and residual access rights
- Risks arising from poor data quality that may adversely impact business decision-making
- Risks of delays and business disruption caused by overly complex data approval processes
- Risks related to insufficient preparedness for data-related incidents
- Risks of PDPA non-compliance and data usage beyond specified purposes

However, if control activities at each stage are not adequately effective, such deficiencies may impact business continuity, stakeholder confidence and the achievement of the Company's strategic objectives. Accordingly, the internal control framework for data governance is defined as follows:

1) Clear data ownership with accountable responsibilities

Data roles are systematically defined, including Data Owner, Data Steward, Data Custodian and Data User, aligned to key data domains such as customer, tenant and employee data. Data sensitivity levels and potential legal and business impacts (e.g., PDPA implications) are assessed accordingly. This approach ensures that all data-related decisions have clearly identified accountability, reduces ambiguity in data usage and mitigates the risk of data misuse. It forms a fundamental pillar of a robust internal data control framework.

2) Role-based access control to mitigate structural risks

Role-Based Access Control (RBAC) is applied as the primary mechanism for managing data access, aligned with job responsibilities on a need-to-know basis. Multi-level approval processes are enforced to enhance control rigor. Access management is segregated between standard (RBAC) and exceptional (Non-RBAC) cases, with periodic access reviews conducted regularly and critical data access reviewed at least annually. These measures help reduce the risk of unauthorized access, prevent systemic errors and enhance the Company's credibility with customers, tenants and business partners.

3) Data protection to support business continuity

Data protection controls are defined across the entire data lifecycle, including data storage, usage, backup, recovery, archiving and proper disposal. A data retention schedule is established in alignment with legal requirements and business needs. Audit logs are maintained, and system recovery or Business Continuity Plans (BCP) are regularly tested. These measures mitigate the risk of data loss, enable effective response to unexpected events and ensure confident compliance with legal obligations.

4) Continuous monitoring of data quality and data incidents to enable effective governance

Data quality monitoring mechanisms are implemented with clear alerting and escalation processes. Data incident management encompasses reporting, investigation, remediation and preventive controls to avoid recurrence. In parallel, key PDPA principles are strictly enforced, including consent management, purpose limitation, and authorized access. This approach ensures

that internal controls extend beyond policy documentation into practical governance execution, thereby strengthening trust among customers, tenants and business partners.

2. REVISION HISTORY

History

No	Revision Date	Division Owner	Revision detail
0	March 2026	BDSI, IT, PDRM	New document
1			
2			

Revisit Plan

ครั้งที่	วันที่ทบทวน	Division Owner	Revision detail
0	March 2027	BDSI , IT	
1			
2			

3. ROLE AND RESPONSIBILITY

3.1) Data Owner

The accountable person for ensuring data is accurate and acceptable for own domain use, approving access rights, defining data classification, and setting policies for the data domain.

3.2) Data Steward

The responsible person for governing data quality and ensuring compliance with data standards signed off from data owner. Define Role based access control matrix for key business data domain with data owner.

3.3) Data Custodian (IT & XPO)

The responsible IT&XPO function for managing data infrastructure, Security controls, Access provisioning according to RBAC Matrix, Data storage, archiving, retention, and disposal, Data backup, recovery, and business continuity. A Data Custodian is responsible for the technical and operational management of data on behalf of the Data Owner.

- Data Security and Protection: Implement and maintain security controls, protect data from unauthorized access, and ensure confidentiality, integrity and availability.

- Access Management: Provision and revoke access based on Data Owner approval and monitor access usage.
- Data Storage and Infrastructure Management: Maintain databases, storage platforms, system performance, patching, and secure configurations.
- Backup, Recovery, and Business Continuity: Ensure backups, disaster recovery and business continuity requirements are met.
- Lifecycle and Data Disposal Management: Execute data retention, archiving and secure disposal.

3.4) Data User

Individuals or teams authorized to use data for operational or business purposes in accordance with approved access rights and policies.

4. PROCESS

4.1 OPERATING PROCESS

4.1.1 THE DESIGNATION OF DATA OWNERS AND DATA DOMAINS (E.G., CUSTOMER, TENANT, HR, AND OTHERS)

This process is used to identify data domains and assess data sensitivity in order to support the assignment of Data Owners, the establishment of appropriate governance mechanisms, and suitable data classification, as follows:

- 1) The Data Steward identifies data domains and data elements by reviewing business processes and system documentation through interviews and data existence assessments, in order to develop a comprehensive data access policy inventory.
- 2) The Data Steward assesses data sensitivity levels and potential legal and business impacts by analyzing PDPA requirements, Company policies, and risk criteria, with reference to the data classification assessment guidelines defined in the Data Classification and Information Management Manual (SPW-LEGAL-MN-003: Data Classification Manual).

4.1.2 THE DEFINITION OF USER ROLES FOR DATA ACCESS (ROLE-BASED ACCESS CONTROL: RBAC)

This process is used to determine data sensitivity levels, approve Role-Based Access Control (RBAC) and communicate defined data domains and access rights to relevant stakeholders as follows:

- 1) The Data Owner determines the data sensitivity level within their respective data domain, reviews the assessment proposals submitted by the Data Steward, and approves the applicable RBAC.
- 2) The Data Steward maintains the RBAC access rights as approved by the Data Owner.

- 3) The Data Steward communicates the defined data domains to relevant teams by publishing the information through governance channels (e.g., the governance portal) or by notifying the IT and business teams for enforcement.

4.1.3 DATA ACCESS PROVISIONING AND MAINTENANCE

1) Data Access Approval - RBAC-Based Access

This process is used to request and provision data access rights based on predefined Role-Based Access Control (RBAC) roles, in order to ensure that access authorization is aligned with assigned responsibilities and the data governance policies. (In this context, RBAC refers to access requests that fall within pre-approved roles and associated access privileges.)

- 1.1) The Data User submits a data access request through the designated system, specifying the required dataset(s), the purpose of use, and the requested access period.
- 1.2) The Data User's direct supervisor reviews and endorses the request to confirm that the requested access is necessary for the assigned job duties.
- 1.3) The Data Steward verifies that the request complies with the existing and approved RBAC roles and is aligned with the defined data domains and data access policies.
- 1.4) The Data Custodian (IT/XPO) provisions data access strictly in accordance with the approved RBAC roles and notifies the Data User upon completion.

2) Data Access Approval - Non-RBAC Access

This process is designed for the request, review, approval, and provisioning of data access rights that fall outside the scope of the existing Role-Based Access Control (RBAC) framework. It covers exceptional, temporary, or ad hoc access and must be justified by business necessity and comply with Company policies and PDPA requirements.

- 2.1) The Data User submits a data access request through the designated system, providing a detailed business justification, the scope of the requested data, and the required access duration.
- 2.2) The Data User's direct supervisor and the next-level supervisor or functional management review and endorse the request to confirm the business necessity prior to the formal approval process.
- 2.3) The Data Steward reviews the endorsed request to validate the necessity, purpose of use, and compliance with PDPA requirements and Company policies, and Data Steward is responsible for coordinating and validating with the Data Owner at least annually.
- 2.4) The Data Owner approves or rejects the request and defines any applicable conditions, scope limitations, or access expiry dates within the approval system.
- 2.5) The Data Steward acknowledges the approved request, updates the RBAC framework where applicable, and notifies the Data Custodian to proceed with access provisioning.
- 2.6) The Data Custodian (IT/XPO) provisions data access strictly in accordance with the Data Owner's approval and notifies the Data User upon completion.

3) Access Rights Maintenance (Business-as-Usual: BAU)

This process is intended to maintain and periodically review data access rights under the Access Rights Maintenance (BAU) process, in order to ensure alignment with user roles, organizational changes, and the Company's data governance policies.

- 3.1) The Data Custodian (IT/XPO) performs ongoing access rights maintenance (BAU) by reviewing, modifying, and revoking access rights in accordance with the Joiner–Mover–Leaver (JML) process.
- 3.2) The Data Steward coordinates with the Data Owner and the Data User's direct supervisor or functional management to periodically review data access rights, or upon organizational changes, to ensure continued alignment with approved RBAC roles and job responsibilities. Access rights to PDPA-related data and confidential data must be reviewed at least once per year.
- 3.3) The Data Owner and the Data User's direct supervisor or functional management review and validate the appropriateness of data access rights for their respective data domains.

4.1.4 QUALITY ESCALATION

This process is used to conduct data quality assessments and report on data quality status in order to enable continuous monitoring and improvement of data quality across the organization.

- 1) The Data Custodian conducts data quality assessments in accordance with the agreed arrangements with the Data Owner and the Data Steward.
- 2) The Data Steward reports material data quality trends and the status of data quality issues through governance meetings or via control dashboards that are made available to the Data Owner and relevant stakeholders.

4.1.5 MAINTAIN DATA DEFINITION & BUSINESS MEANING

This process is intended to maintain and enhance data definitions and business meanings in order to ensure that data definitions, business meanings, and system-level data attributes remain accurate and up to date at all times.

- 1) The Data Steward maintains the data definitions and business meanings by adding or revising definitions, business rules, and data ownership details within the data catalog.
- 2) The Data Custodian maintains technical data definition and business meaning information by recording data structures, data records and system-level attributes through data definition and business meaning management tools, or through automated alignment mechanisms.

4.1.6 DATA STORAGE, ARCHIVING AND RETENTION/DISPOSAL

This process is intended to manage the secure retention, archiving, and disposal of data in accordance with the data retention schedule and the requirements set forth in the Company's data retention policies and guidelines under the Siam Piwat Group.

- 1) The Data Owner defines data retention periods by reviewing applicable regulatory requirements and business needs, and records the required retention periods in the Data Retention Schedule.
- 2) The Data Custodian identifies data that meets the criteria for archiving or disposal by querying systems in accordance with the defined data retention schedule.
- 3) The Data Custodian performs secure data archiving or disposal by transferring data to an archive storage area or applying certified data deletion methods, and retains supporting evidence in audit logs.

Note: The data platform is currently under development.

4.1.7 DATA BACK-UP, RECOVERY AND BUSINESS CONTINUITY

This process is intended to ensure regular data backup and recovery testing in order to maintain data availability and support business continuity requirements.

- 1) The Data Custodian performs scheduled data backups by executing backup activities on designated systems in accordance with the established Backup and Restore Procedure.
- 2) The Data Custodian conducts recovery testing (recovery drills) by restoring systems within the disaster recovery (DR) environment in accordance with the Information Technology Business Continuity Plan (IT BCP).

4.1.8 DATA SECURITY AND PROTECTION

This process is intended to implement and enforce technical security controls to protect data in compliance with the Company's Information Technology Security Policy (IT Security Policy).

- 1) The Data Custodian implements and operates technical security controls in accordance with the IT Security Policy.
- 2) The Data Custodian enforces access controls, data encryption, event logging, monitoring, and other data protection mechanisms.

4.2 INCIDENT MANAGEMENT

This process is intended for the reporting, investigation, and management of data-related incidents in order to ensure timely response, remediation, and prevention of recurrence, in accordance with the Incident Management Procedure. Data-related incidents include, but are not limited to, data breaches, unauthorized access, data loss or corruption, and data integrity failures.

- 1) All data incidents must be reported and recorded through the designated channels in accordance with the Incident Management Procedure.
- 2) The Data Steward and the Data Custodian investigate incidents and carry out incident containment, recovery actions, and technical remediation.
- 3) The Data User provides support for impact assessment and validates the accuracy and effectiveness of corrective measures.
- 4) The Data Custodian retains technical evidence, supports root cause analysis, and implements corrective and preventive measures to prevent recurrence.

4.3 COMPLIANCE AND PDPA REQUIREMENTS

All activities related to data management must comply with the Personal Data Protection Act (PDPA) and the Company's internal governance requirements, including the following:

- 1) Consent must be recorded when personal data is collected or used.
- 2) Personal data must be used solely for the specified and approved purposes.

Process Risk and Control Table

Process	Risk	Control
1) The designation of Data Owners and data domains	- Incomplete or overlapping identification of data domains. - Incorrect assessment of data sensitivity. (PDPA/ confidential data) - Inability to clearly identify accountable data owners.	- The Data Steward reviews business processes and system documentation through interviews and data discovery. - Data sensitivity levels are assessed with reference to PDPA requirements, Company policies, and the data classification manual.
2) Data Access Approval	- Excessive data access beyond business necessity. - Approvals that are not aligned with data classification requirements. - Lack of documented evidence of approval.	- Submission of access requests through the designated system. - The Data Owner reviews the business necessity, purpose of use, and alignment with data classification. - Recording the approval outcome and associated conditions within the system.
3) User Role Definition (RBAC)	- RBAC roles are not aligned with data sensitivity levels. - Access rights are not up to date. - Relevant stakeholders are not aware of the defined data	- The Data Owner approves data sensitivity levels and RBAC roles. - The Data Steward maintains access rights in accordance with RBAC.

Process	Risk	Control
	domains and assigned access rights.	- Communication is conducted through the governance portal, with notifications provided to IT and business teams.
4) RBAC-Based Data Access	- Access requests that do not align with the actual user roles. - Incorrect provisioning of access rights due to misalignment with the approved RBAC. - Unclear segregation of duties.	- Endorsement by the direct supervisor. - The Data Steward verifies alignment with RBAC. - The Data Custodian provisions access rights strictly in accordance with RBAC.
5) Non-RBAC Data Access	- Excessive use of exceptions beyond business necessity. - Access to PDPA-related data without appropriate justification. - Temporary access rights are not revoked in a timely manner.	- Excessive use of exceptions beyond business necessity. - Access to PDPA-related data without appropriate justification. - Temporary access rights are not revoked in a timely manner.
6) Access Rights Maintenance (BAU)	- Employees who have resigned or changed roles retain access rights. - PDPA-related access rights are not reviewed in accordance with the defined review cycle.	- Execution of the Joiner-Mover-Leaver (JML) process. - Periodic access rights reviews jointly conducted by the Data Steward, Data Owner and functional management. - Requirement that access to related data be reviewed at least once per year.
7) Data Quality Escalation	- Data quality issues are not detected. - There is no monitoring or tracking of data quality trends.	- The Data Custodian conducts data quality checks in accordance with agreed arrangements. - The Data Steward reports through governance meetings and dashboards.
8) Data Definition Maintenance	- Data definitions are inconsistent between business and IT.	- The Data Steward manages and maintains business definitions within the data catalog.

Process	Risk	Control
	- Data lineage is inaccurate or not up to date.	- The Data Custodian maintains technical metadata
9) Data Archiving, Retention and Disposal	- Data is retained beyond the legally prescribed retention period. - Data is disposed of without supporting evidence. - Critical data is erroneously destroyed.	- The Data Owner defines the data retention schedule. - The Data Custodian identifies data in accordance with the defined retention criteria. - Evidence of data archiving and disposal is recorded in audit logs.
10) Data Backup and Business Continuity	- Data cannot be recovered when an incident occurs. - The disaster recovery (DR) plan is not operationally ready for actual use.	- Data backups are performed in accordance with the Backup and Restore Procedure. - Recovery drills are conducted in accordance with the Business Continuity Plan (BCP).
11) Data Security and Protection	- Data leakage or unauthorized access to data. - Non-compliance with the IT Security Policy.	- Enforcement of technical controls by the Data Custodian. - Data encryption, logging and monitoring.
12) Incident Management	- Data incidents are not reported or are reported with delays - Root cause analysis is not conducted	- Reporting in accordance with the Incident Management Procedure. - Investigation, recovery activities and evidence preservation performed by the Data Custodian.
13) Compliance & PDPA	- Use of personal data beyond the specified purpose. - Lack of documented consent.	- Recording of consent. - Ensuring that data is used solely for approved and specified purposes.

5. APPENDIX AND REFERENCE DOCUMENT

- 1) SPW-LEGAL-MN-003: Data Classification and Information Management Manual of the Siam Piwat Group (Data Classification Manual)
- 2) The Siam Piwat Group's Policy and Guidelines on Corporate Data Retention
- 3) The Company's Information Technology Security Policy (IT Security Policy)
- 4) Roles and Responsibilities Matrix

Processes / Activities	Data Owner	Data Steward	Data Custodian (IT/XPO)	Data User
1. Assignment of Data Owners and Data Domains	A	R	C	I
2. Definition of Data User Roles (RBAC)	A	R	C	I
3. Provisioning and Maintenance of Data Access	A	C	R	I
4. Data Quality Issue Management	A	C, R	C, R	I
5. Maintenance of Data Definitions and Business Meanings	C	A, R	C	I
6. Data Storage, Archiving and Disposal	A	C	R	I
7. Data Backup, Recovery and Business Continuity Planning (BCP)	A	C	R	I
8. Data Security and Protection	A	C	R	I

Note:

R - Responsible:

The individual(s) who perform the work and are directly responsible for executing the task. This role owns the operational activities and must ensure that the work is completed satisfactorily.

A - Accountable:

The individual who has ultimate ownership of the outcome and decision-making authority. This role is accountable for the overall success of the task or activity. Each task or activity should have only one Accountable.

C - Consulted:

The individual(s) who provide input, information, or expert advice prior to execution, such as subject matter experts or impacted stakeholders. Communication is two-way.

I - Informed:

The individual(s) who must be kept informed of progress or outcomes but are not required to be involved in decision-making. Communication is one-way.

Based on the Roles and Responsibilities matrix, the details can be described as follows:

1) Data Owner

The Data Owner plays a primary role in defining the direction, standards and policies related to data within the assigned data domain. The responsibilities include the following:

- Defining data domains: Identifying the Data Owner and associated responsibilities for each data domain.

- Approving data access rights: Reviewing business appropriateness and security considerations.
- Defining user roles (RBAC): Establishing the data access structure through Role-Based Access Control.
- Addressing data quality issues: Taking responsibility at the policy and governance level.
- Managing data across its lifecycle: Overseeing data storage, archiving and disposal.
- Supporting data backup and recovery: Ensuring business continuity readiness (BCP).
- Ensuring data security: Being accountable for protecting data within the assigned domain.

In addition, the consultative role of the Data Owner involves providing business guidance to the Data Steward in the maintenance of data definitions and business meanings.

2) Data Steward

The Data Steward is responsible for overseeing data management and data quality throughout the data lifecycle. This role focuses on maintaining data quality and standards, with the following responsibilities:

- Defining Data Owners: Identifying data domains and the corresponding roles and responsibilities.
- Approving data access: Exercising governance oversight over data access approvals.
- Provisioning data access: Collaborating with the Data Custodian and providing support and coordination with relevant functions to ensure that data access rights are reviewed at least once per year.
- Addressing data quality issues: Working jointly with the Data Custodian to resolve data quality matters.
- Maintaining business meanings and data definitions: Ensuring that data meanings and definitions are accurate and consistent.
- Managing data lifecycle practices: Providing advisory guidance on data storage, archiving and disposal.
- Overseeing data security: Providing recommendations and guidance on data protection measures.

3) Data Custodian (IT/XPO)

The Data Custodian (IT/XPO) is responsible for technical oversight of data storage and data protection. This role serves as the technical backbone to ensure the stability and reliability of the data infrastructure. The responsibilities include the following:

- Defining Data Owners and data domains: Providing technical advice and recommendations.
- Approving access rights and user roles: Collaborating with the Data Steward.
- Provisioning data access: Being responsible for granting access in accordance with established guidelines.
- Addressing data quality issues: Working jointly with the Data Steward to resolve data quality problems.

- Maintaining business meanings: Providing technical guidance related to business meanings and definitions.
- Managing data lifecycle activities: Overseeing data archiving and data disposal processes.
- Ensuring data security: Being responsible for system- and technology-level data protection.

4) Data User

A Data User utilizes data in accordance with the access rights granted. This role is critical to ensuring that data is used correctly and securely, with the following responsibilities:

- Awareness of data domains: Understanding the applicable access approval processes and assigned roles.
- Use of data in accordance with granted access rights: Utilizing data appropriately and in compliance with Company policies.
- Awareness of data quality issues: Being informed of data quality issues and any remediation actions that may affect data usage.
- Understanding of data meanings: Being knowledgeable about the maintenance of data definitions and business meanings.
- Awareness of data management processes: Understanding data storage, archiving, and disposal processes.
- Compliance with security standards: Acknowledging and complying with data security policies and standards.